

Lab Title: Firewall Traffic Control and Forensic Verification

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**Course Name: SBT-DF203: Basic
Networking Skills for Digital Forensics**

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International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 6: Firewall Traffic Control and Forensic Verification

Three-Week Delivery Block 2/3 | 12-18 September 2026

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Important Pre-Lab Instruction Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.	
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 6
Lab Title	Firewall Traffic Control and Forensic Verification
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	4-5 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	Baseline and blocked HTTP PCAPNG files; iptables ruleset exports

1. Source Materials and Slide Alignment

- Firewall: network-based and host-based firewalls, iptables chains and rules, ACCEPT/DROP actions, UFW relationship, NFQUEUE concept and HTTP blocking.
- The slide practical lists rules, adds/removes an INPUT rule, uses a local Apache page and observes allowed versus blocked hosts.
- The homework requires traffic generated by curl from a different machine and forensic observation of the result.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are validating whether a Linux host-based firewall correctly blocks HTTP traffic from one authorized lab client while allowing another. You will capture baseline traffic, apply a narrowly scoped iptables rule, capture the blocked connection behaviour, prove the rule hit using counters, remove the rule and document the evidential differences.

3. Learning Outcomes

- Explain the purpose of host-based and network-based firewalls.
- Interpret INPUT, OUTPUT and FORWARD chains.
- Create, verify and remove a narrowly scoped iptables rule.
- Capture allowed and blocked HTTP connection attempts.
- Correlate packet evidence with firewall rule counters and application behaviour.
- Explain DROP versus REJECT and identify retransmission/timeout evidence.
- Restore the original ruleset safely.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Before making changes, export the existing ruleset. Apply only a source-specific rule for the assigned lab IP. Never flush all firewall rules on a remotely administered or production system.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Server VM	Kali/Ubuntu with Apache	Protected host.
Client VM	Linux or Windows on host-only network	Generate HTTP requests.
Firewall tools	iptables, iptables-save	Apply and preserve rules.
Capture tools	tshark/Wireshark	Capture baseline and blocked traffic.
Client tools	curl, browser	Test access.
Optional extension	NetfilterQueue	Instructor-led packet interception demonstration.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab6
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y apache2 curl iptables tshark wireshark
sudo systemctl enable --now apache2
printf '<!DOCTYPE html>\n<html><body><h1>ICDFA Network Forensics Firewall Lab</h1><p>Server: YOUR SERVER NAME</p></body></html>\n' | sudo tee /var/www/html/firewall_lab.html
sudo iptables-save | tee reports/iptables_before.rules
sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
sha256sum reports/iptables_before.rules | tee reports/iptables_before_sha256.txt
```

```
kali@kali: ~/SBT-DF203-Lab6

Session Actions Edit View Help

(kali㉿kali)-[~]
$ mkdir -p ~/SBT-DF203-Lab6/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/SBT-DF203-Lab6

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ pwd
/home/kali/SBT-DF203-Lab6

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ sudo apt install -y apache2 curl iptables tshark wireshark
[sudo] password for kali:
apache2 is already the newest version (2.4.68-1).
curl is already the newest version (8.21.0-2).
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gdal-data libgeotiff5 libmuparser2v5 onboard-common
  gdal-plugins libgirepository-1.0-1 libnetcdf2 proj-bin
  gir1.2-ayatanaappindicator3-0.1 libgpgme45 libodbc2 proj-data
  gir1.2-girepository-2.0 libhdf4-0 libodbcrc2 python3-fs
  libaec0 libhdf5-310 libodbcinst2 python3-gpg
  libarmadillo14 libhdf5-hl-310 libportmidi0 python3-talloc
  libarpack2t64 libjs-jquery-ui libproj25 python3-tdb
  libavc1394-0 libjs-underscore libpython3.13-dev python3.13
  libblosc2-4 libjxr-tools libraw23t64 python3.13-dev
  libcfitsio10t64 libjxr0t64 librttopo1 python3.13-minimal
  libcrypt-dev libkmlbase1t64 libspatialite8t64 python3.13-venv
  libfreexl1 libkmlengine1t64 libwireshark18 rpcsvc-proto
  libfyba0t64 libkmlengine1t64 libwiretap15 samba-ad-provision
  libgav1-1 libmbedcrypto16 libwmflite-0.2-7 samba-dsdb-modules
```

```

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 10:09:10 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-ISOLATION-STAGE-1
-A FORWARD -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A FORWARD -o docker0 -j DOCKER
-A FORWARD -i docker0 ! -o docker0 -j ACCEPT
-A FORWARD -i docker0 -o docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-1 -j RETURN
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
-A DOCKER-ISOLATION-STAGE-2 -j RETURN
-A DOCKER-USER -j RETURN
COMMIT
# Completed on Fri Sep 18 10:09:10 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Fri Sep 18 10:09:10 2026
*nat
:PREROUTING ACCEPT [2:656]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [157:13418]
:POSTROUTING ACCEPT [157:13418]
:DOCKER - [0:0]
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
-A DOCKER -i docker0 -j RETURN
COMMIT
# Completed on Fri Sep 18 10:09:10 2026

```

```

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_before.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DOCKER-USER all -- *      *       0.0.0.0/0             0.0.0.0/0
2    0    0 DOCKER-ISOLATION-STAGE-1 all -- *      *       0.0.0.0/0             0.0.0.0/0
3    0    0 ACCEPT    all -- *      docker0 0.0.0.0/0             0.0.0.0/0
4    0    0 DOCKER    all -- *      docker0 0.0.0.0/0             0.0.0.0/0
5    0    0 ACCEPT    all -- docker0 !docker0 0.0.0.0/0             0.0.0.0/0
6    0    0 ACCEPT    all -- docker0 docker0  0.0.0.0/0             0.0.0.0/0

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain DOCKER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DOCKER-ISOLATION-STAGE-2 all -- docker0 !docker0 0.0.0.0/0             0.0.0.0/0
2    0    0 RETURN    all -- *      *       0.0.0.0/0             0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DROP     all -- *      *       0.0.0.0/0             0.0.0.0/0
2    0    0 RETURN    all -- *      *       0.0.0.0/0             0.0.0.0/0

Chain DOCKER-USER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 RETURN    all -- *      *       0.0.0.0/0             0.0.0.0/0

```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab6-Samuel_Chinedu_Onu
Trainee name	Samuel Chinedu Onu
Date and time started	18th September 2026, 9:15am WAT
Evidence file name(s)	http_allowed.pcapng, http_blocked.pcapng + iptables ruleset exports
Source or generation method	Locally generated in isolated lab environment using Apache, iptables, curl and tshark on Kali linux VM
Original SHA-256	bcac92d37bb9a2d775b7411050220cee8f6709e9ac83a541e36f6ac3b762a873
Working-copy SHA-256	
Analysis workstation/VM	Kali Linux VM
Notes on any changes	Original firewall rules were exported before any modification

8. Part A - Record the Lab Network and Baseline Access

```
# On the server
ip -br address | tee reports/server_interfaces.txt
ip route | tee reports/server_routes.txt
sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt

# On the client, replace SERVER_IP
curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html 2>&1 | tee baseline_curl.txt
```

Record the server IP, allowed client IP, blocked client IP, interface and default gateway. Take a snapshot before changing the firewall.

```
kali@kali: ~/SBT-DF203-Lab6
Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab6]
$ ip -br address | tee reports/server_interfaces.txt
lo                UNKNOWN      127.0.0.1/8  ::1/128
eth0              UP          192.168.225.128/24  fe80::92cd:feed:c2d:71f9/64
eth1              UP          192.168.88.130/24  fe80::afe1:5436:8b42:335d/64
docker0          DOWN        172.17.0.1/16

(kali@kali)-[~/SBT-DF203-Lab6]
$ ip route | tee reports/server_routes.txt
default via 192.168.225.2 dev eth0 proto dhcp src 192.168.225.128 metric 102
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.88.0/24 dev eth1 proto kernel scope link src 192.168.88.130 metric 101
192.168.225.0/24 dev eth0 proto kernel scope link src 192.168.225.128 metric 102

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo ss -lntp | grep ':80' | tee reports/apache_listener.txt
[sudo] password for kali:
LISTEN 0      511          *:80          *:~          users:((("apache2",pid=1086,fd=4),("apache2",pid=1085,fd=4),("a
pache2",pid=1084,fd=4),("apache2",pid=1083,fd=4),("apache2",pid=1082,fd=4),("apache2",pid=996,fd=4)))

(kali@kali)-[~/SBT-DF203-Lab6]
$ # On the client, replace SERVER_IP
curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html 2>&1 | tee baseline_curl.txt
* Resolving timed out after 5002 milliseconds
* closing connection #0
curl: (28) Resolving timed out after 5002 milliseconds

(kali@kali)-[~/SBT-DF203-Lab6]
$
```

9. Part B - Capture the Allowed HTTP Baseline

```
# On server, replace IFACE and CLIENT_IP
sudo tshark -i IFACE -f 'host CLIENT_IP and tcp port 80' -a duration:30 -w evidence/http_allowed.pcapng &
sleep 3
# Run curl from client during this capture
wait
sha256sum evidence/http_allowed.pcapng | tee reports/http_allowed_sha256.txt
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ # On server, replace IFACE and CLIENT_IP
sudo tshark -i IFACE -f 'host CLIENT_IP and tcp port 80' -a duration:30 -w evidence/http_allowed.pcapng &
sleep 3
[1] 39097
[sudo] password for kali:
[1] + suspended (tty input) sudo tshark -i IFACE -f 'host CLIENT_IP and tcp port 80' -a duration:30 -w

(kali㉿kali)-[~/SBT-DF203-Lab6]
$ # Run curl from client during this capture
wait
```

10. Part C - Apply and Verify the Blocking Rule

Use the exact source IP of the assigned blocked client. Insert the rule near the top of INPUT so it is evaluated before a broad ACCEPT rule.

```
BLOCKED_CLIENT_IP=192.168.56.20
sudo iptables -I INPUT 1 -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_add.txt
sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
printf 'Rule verified present.\n' | tee reports/rule_verification.txt
```

11. Part D - Capture Blocked Traffic and Rule Counters

```
# On server
sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &
sleep 3
# From blocked client during capture:
# curl -v --connect-timeout 10 http://SERVER_IP/firewall_lab.html
wait

sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
```

```

Session Actions Edit View Help

(kali@kali)-[~/SBT-DF203-Lab6]
$ # On server
sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -w evidence/http_blocked.pcapng &
sleep 3
[1] 4098
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'IFACE'
tshark: There is no device named "IFACE".
(No such device exists)
[1] + exit 1      sudo tshark -i IFACE -f 'host 192.168.56.20 and tcp port 80' -a duration:35 -

(kali@kali)-[~/SBT-DF203-Lab6]
$ # From blocked client during capture:
# curl -v --connect-timeout 10 http://SERVER_IP/firewall_lab.html
wait

(kali@kali)-[~/SBT-DF203-Lab6]
$ sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_after_test.txt
Chain INPUT (policy ACCEPT 4 packets, 252 bytes)
num  pkts bytes target    prot opt in     out     source               destination
1      0    0 DROP      tcp  --  *      *       192.168.56.20        0.0.0.0/0           tcp dpt:80

(kali@kali)-[~/SBT-DF203-Lab6]
$ sha256sum evidence/http_blocked.pcapng | tee reports/http_blocked_sha256.txt
sha256sum: evidence/http_blocked.pcapng: No such file or directory

(kali@kali)-[~/SBT-DF203-Lab6]
$

```

12. Part E - Compare Allowed and Blocked Captures

```

for PCAP in evidence/http_allowed.pcapng evidence/http_blocked.pcapng; do
  echo "===== $PCAP ====="
  tshark -r "$PCAP" -Y 'tcp.flags.syn==1 || http.request || http.response || tcp.analysis.retransmission' \
    -T fields -e frame.number -e frame.time_relative -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport \
    -e tcp.flags -e tcp.analysis.retransmission -e http.request.uri -e http.response.code
done | tee reports/allowed_vs_blocked.tsv

```

Indicator	Allowed Capture	Blocked Capture
Client SYN visible?	Yes	Yes
Server SYN-ACK visible?	Yes	No
Handshake completed?	Yes	No
HTTP GET visible?	Yes	No
HTTP response visible?	Yes	No
Retransmissions/timeouts	Yes	Yes
iptables counter change	No change	There was an increase
curl result	successful	Times out

13. Part F - Remove the Rule and Restore Access

```
BLOCKED_CLIENT_IP=192.168.56.20
sudo iptables -D INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP || echo 'Rule successfully removed'
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
# Confirm access from the previously blocked client
# curl -v --connect-timeout 5 http://SERVER\_IP/firewall\_lab.html
```

```
(kali㉿kali)-[~/SBT-DF203-Lab6]
$ BLOCKED_CLIENT_IP=192.168.56.20
sudo iptables -D INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP
sudo iptables -C INPUT -s "$BLOCKED_CLIENT_IP" -p tcp --dport 80 -j DROP || echo 'Rule successfully removed'
sudo iptables -L INPUT -n -v --line-numbers | tee reports/iptables_restored.txt
# Confirm access from the previously blocked client
# curl -v --connect-timeout 5 http://SERVER_IP/firewall_lab.html

iptables: Bad rule (does a matching rule exist in that chain?).
Rule successfully removed
Chain INPUT (policy ACCEPT 8 packets, 808 bytes)
num  pkts bytes target    prot opt in     out     source                   destination

(kali㉿kali)-[~/SBT-DF203-Lab6]
$
```